3.9 Graphing Piecewise Functions

Name _____ Period ____ Date _____

Algebra I 3.9 · Student sheet

I can…

- evaluate piecewise functions.
- graph piecewise functions.
- write and use piecewise functions for real-life situations.

Vocabulary · fill in the missing word

Piecewise function — a function _____ by two or more rules, each on its own part of the domain

Open dot — the _____ is not included

Closed dot — the _____ is included

Part 1 · Parking garage

$5 for the first hour and $3 for each additional hour. The most you pay in a day is $18.

Hours	1	2	3	4	5	6
Cost without a maximum						
Cost with the $18 maximum						

Where does the rule change? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Piecewise function: A piecewise function uses _____ rules, each on its own part of the _____.

Evaluating: To evaluate, first decide which _____ the input belongs to.

Example 1

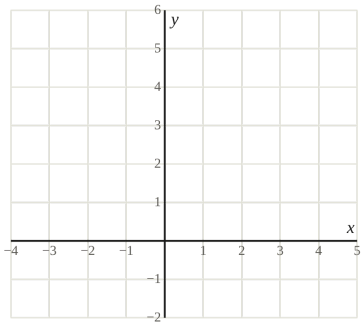
$f(x) = 2x + 1$ if $x < 0$; $f(x) = x^2$ if $x \geq 0$.

Find $f(-3)$, $f(0)$, and $f(4)$.

Open and closed dots: Open dot: _____. Closed dot: _____.

Example 2

Graph $f(x) = x + 2$ if $x < 1$; $f(x) = -x + 5$ if $x \geq 1$.



Example 3

A phone plan costs \$40 for up to 5 GB of data and \$10 for each GB over 5:

$C(g) = 40$ if $g \leq 5$; $C(g) = 40 + 10(g - 5)$ if $g > 5$.

What is the cost for 7 GB?

Part 3 · Partner practice

Partner 1

$f(x) = -x$ if $x \leq 2$; $f(x) = 3x - 4$ if $x > 2$.

Find $f(2)$ and $f(5)$.

Partner 2

Use the phone plan from Example 3. What is the cost for 3 GB?